

16. (3 points)

Given the following information:

Accident Year	Rate-Level Adjusted Premium (\$000)	Aggregate Reported Loss at Dec. 31, 2016 (\$000)	Aggregate Loss Report Lag at Dec. 31, 2016	Aggregate Reported Loss at Jun. 30, 2017 (\$000)	Aggregate Loss Report Lag at Jun. 30, 2017
2014	6,000	3,000	0.70	3,500	0.75
2015	7,000	2,000	0.50	2,200	0.55
2016	10,000	4,000	0.30	4,800	0.35
Total	23,000	9,000	N/A	10,500	N/A

- The selected credibility factor is 0.50.

a. (2.5 points)

Test the total expected claims emergence from December 31, 2016 to June 30, 2017 against the total actual claims emergence using a credibility-weighted combination of the Stanard-Bühlmann and chain ladder methods.

b. (0.5 point)

Provide two reasons which would explain the difference calculated in part a. above.

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SPRING 2018 EXAM 7, QUESTION 16

TOTAL POINT VALUE: 3

LEARNING OBJECTIVE(S): A13

SAMPLE ANSWERS

Part a: 2.5 points

Sample 1

AY	Reported Loss	Used-Up Premium	SB IBNR	CL IBNR
2014	3,000	$6,000 \times 0.7 = 4,200$	$6,000 \times 0.8411 \times (1-0.7) = 1,513.98$	$3000 / 0.7 - 3,000 = 1,285.71$
2015	2,000	$7,000 \times 0.5 = 3,500$	$7,000 \times 0.8411 \times (1-0.5) = 2,943.85$	$2000 / 0.5 - 2,000 = 2,000$
2016	4,000	$10,000 \times 0.3 = 3,000$	$10,000 \times 0.8411 \times (1-0.3) = 5,887.7$	$4000 / 0.3 - 4,000 = 933.33$
Total	9,000	10,700	ELR = 9,000 / 10,700 = 0.8411	

AY	Z	Cred IBNR	Expected Emergence
2014	$0.5 \times 0.7 = 0.35$	$0.35 \times 1,285.71 + (1 - 0.35) \times 1,513.98 = 1,434.09$	$1,434.09 \times (0.75 - 0.7) / (1 - 0.7) = 239.02$
2015	$0.5 \times 0.5 = 0.25$	$0.25 \times 2000 + (1 - 0.25) \times 2,943.85 = 2,707.89$	$2,707.89 \times (0.55 - 0.5) / (1 - 0.5) = 270.79$
2016	$0.5 \times 0.3 = 0.15$	$0.15 \times 9,333.33 + (1 - 0.15) \times 5,887.7 = 6,404.54$	$6,404.54 \times (0.35 - 0.3) / (1 - 0.3) = 457.47$

AY	Actual Emergence	Difference
2014	$3,500 - 3,000 = 500$	$500 - 239.02 = 260.98$
2015	$2,200 - 2,000 = 200$	$200 - 270.79 = -70.79$
2016	$4,800 - 4,000 = 800$	$800 - 457.47 = 342.53$
Total		532.72

Actual emergence higher than expected emergence

Sample 2

AY	(1) Adj Prem	(2) Report Lag	(3)=(1)×(2) Used-up premium	(4) Reported Losses	(5) Reported Lag at 6/30/2017
2014	6,000	0.7	4,200	3,000	0.75
2015	7,000	0.5	3,500	2,000	0.55
2016	10,000	0.3	3,000	4,000	0.35
			10,700	9,000	

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$(6)=(5)-(2)$	$(7)=(6)\times(1)\times\text{ELR}$	$(8)=(4)/(2)\times(6)$	$(9)=(2)\times 0.5$	$(10)=(9)\times(8) + [1-(9)]\times(7)$
Lag Difference	Expected Rep Loss using CC	Expected Rep Loss using CL	z	Expected IncLoss Cred
0.05	252	214	0.35	239
0.05	294	200	0.25	271
0.05	421	667	0.15	458
				968
Expected Loss Emerged = 968 Actual Loss Emerged = 10,500 – 9,000 = 1,500 Expected – Actual = -532				
Part b: 0.5 point				
<u>Sample responses:</u> • Our estimate was too low so perhaps our initial IBNR was too low. • Maybe it's just random chance. • The company begins a new monitoring program that increases how fast claims are reported to the company. • The company increases the average case reserves due to a management change. • The claim report lag is too short. • We may have had a large loss. • Case load effect: losses are being reported faster than usual because there are fewer claims per claims staff member. • Legislative change: law might have shortened the statute of limitations on a claim so claimants are reporting faster than before the lag was set.				
EXAMINER'S REPORT				
Candidates were expected to know how to calculate expected loss emergence and why the actual emergence could be different than expected.				
Part a				
Candidates were expected to know how to calculate expected loss emergence using both Chain-Ladder and Standard-Bühlmann methods, and how to credibility weight the two methods.				
Common mistakes included: • Not multiplying the credibility factor by the lags. • Using the wrong evaluation date values. • Incorrectly applying credibility to CL and SB.				
Part b				
Candidates were expected to understand why the actual emergence could be different than expected.				
Common mistakes included: • Providing only one reason.				

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

- Simply stating that losses developed more than expected.
- Claiming that the credibility factor was too low without justification. If part a. is done correctly, the actual is significantly higher than expected for either of the methods.